INCH-POUND

A-A-59829
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COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, FORK, ELECTRIC, SIT DOWN, SOLID TIRES, 12000 POUND CAPACITY AT 24 INCH LOAD CENTER, 184 INCHES MINIMUM LIFT HEIGHT

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE

1.1 Description. This commercial item description covers the general requirements for an electric, front wheel drive, rear-wheel steer, sit down operator, solid tired, 12,000 pound at 24 inch load center capacity, forklift truck. It will be Type EE, unless otherwise specified as Type E, in accordance with UL 583. It will be used for handling cargo in and around warehouses, loading platforms, docks, and on paved or other hard surfaces.

2. SALIENT CHARACTERISTICS

2.1 Safety. The forklift shall comply with ANSI/ITSDF B56.1, UL 583, and OSHA standards in effect at the time of manufacture. The forklift shall comply with UL 583, EE rated. When specified (see 7.3), the forklift shall comply with UL 583, E rated. A commercial fire extinguisher shall be mounted on the truck in an easily accessible location. The fire extinguisher shall be a minimum 2-1/2 pound capacity ABC type, or equivalent. The fire extinguisher shall be UL listed. An operator’s seatbelt conforming to SAE J386, and restraint devices designed to ensure the operator's upper body remains entirely within the protection of the overhead guard in the event of tipover shall be provided. Unless otherwise specified (see 6.3), the final color shall be yellow for safe operation in an indoor environment.

2.2 Environmental requirements. The truck shall be capable of operating in temperatures ranging from 0 degrees F to +125 degrees F.

2.3 Battery. The battery shall be a commercially available 48, 72, or 80 volt battery and 1600 amp hours. The battery-mounted, half connector shall be “SB” type. When specified (see 6.3), an additional battery and/or the appropriate battery charger shall be shipped with the vehicle.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: 642 CBSG/GBEC,460 Richard Ray Blvd; Suite 200 Robins AFB, GA 31098-1813.

AMSC N/A FSC 3930

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2.4 Hydraulic system(s). A pressure relief protection device is required along with pump(s), cylinders, control valves, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system. The hydraulic system shall also be designed to prevent forward tilt in the event of system failure; tilt angle shall not deviate more than two degrees if failure occurs at any position during load handling operations. The hydraulic lift system shall be designed so that in the event of hydraulic hose or pump failure, the load shall not descend (freefall) at an uncontrolled rate; lowering speed shall not exceed eight (8) inches per second under any load condition. A manual override for the tilt function shall also be provided to allow for load removal.

2.5 Uprights and carriage. With no load on the forks, the seated operator with seatbelt engaged shall be able to see at least one fork tip at all lift heights and fork positions.

2.6 Forks and carriage. Fork tines shall have the following standard dimensions. The forks shall be mounted so that when fully lowered and forklift operating in reverse, there is no damage or degradation to the fork tines, their attachment, or any other component of the carriage, including fork positioner and side shift. When specified (see 6.3), optional forks shall be furnished.

Standard Forks
a. Length: 48 ± 1 inches
b. Width: 6 inches (maximum)
c. Thickness: 2.5 inches maximum

Optional Forks
a. Length: 72 ± 1 inches
b. Width: 6 inches (maximum)
c. Thickness: 2.5 inches (maximum)

2.7 Fork positioner. A hydraulically powered fork positioner shall be furnished and controlled by the operator while seated. The fork positioner shall allow tine spacing of 18 inches (or less) to 60 inches (or more), measured between centerlines of the forks.

2.8 Side shift. Hydraulically powered side shift capability shall be furnished. It shall be controlled by the seated operator, and shall provide at least six inch side shift capability (each side of center or 12.0 inches minimum total travel).

2.9 Load backrest. The load backrest, in conjunction with the forks and carriage, shall provide a vertical rear load guard of at least 48 inches high, measured from the load carrying surfaces of the forks.

2.10 Steering. Power steering, with emergency steering in the event of power failure, shall be furnished.

2.11 Service brakes. Power brakes conforming to ANSI/ITSDF B56.1 shall be furnished.

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2.12 Parking brake. Parking brake shall conform to ANSI/ITSDF B56.1.

2.13 Electrical system. A 48, 72, or 80 volt direct current (VDC) electrical system shall be furnished. The electrical system shall consist of all electrical components necessary for operation of the forklift. The forklift shall have sufficient electrical grounding to prevent static discharge.

2.14 Instruments and controls. In addition to the instruments supplied on the standard commercial forklift, an hour meter shall be installed in the instrument panel. Except the hour meter, all instruments shall illuminate. All load motion controls shall be right hand operation controls and of the self-centering type, i.e., controls shall return to the neutral position when released.

2.15 Lighting. Minimum two sealed beam floodlights, one facing forward and one facing rearward, shall be furnished. Tail light(s) and brake stop light(s) shall be installed. All lights shall be protected by location or guards. Individual operator controlled switches shall be provided for front floodlight, and rear floodlight.

2.16 Horn. Horn button shall be mounted in the center of the steering wheel.

2.17 Operator's overhead guard. The guard shall be in accordance with ANSI/ITSDF B56.1. It shall also be a falling-object-protective-structure (FOPS) conforming to ANSI/ITSDF B56.1. Overhead guard height from ground to top of guard shall be 102 inches maximum.

2.18 Cab. When specified (see 6.3), the forklift, shall be equipped with an enclosed cab that shall incorporate FOPS requirements of ANSI/ITSDF B56.1. The cab shall have a door on each side of the vehicle with at least one window per door which can be opened for ventilation; the doors shall be capable of being locked in either the open or closed positions. In addition to the door windows, the cab shall be furnished with a front windshield, rear window, and roof window. The windshield and all windows shall be safety glass. The cab interior shall be furnished with water-resistant non-absorbent thermal insulation. A commercial fire extinguisher shall be mounted on the truck in an easily accessible location. The fire extinguisher shall be a minimum 2-1/2 pound capacity ABC type, or equivalent. The fire extinguisher shall be UL listed. The manufacturer's heaviest duty heater and defroster shall be furnished. There shall be at least two wipers furnished, one on the front windshield and one on the rear window.

2.19 Tires. Tire loading shall not exceed those specified in the Tire and Rim Association Yearbook.

2.20 Performance.

a. Capability to safely handle a 12,000 pound, 48 inch per side cube with the center of gravity at the center of the cube. It shall be able to handle this load at all lift heights that the forklift is capable of attaining.
b. Lifting speed. At least 7.0 inches per second with rated load on the forks.

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c. Lowering speed. Not more than 18 inches per second with rated load on the forks. Not less than 6 inches per second with no load and with engine at idle speed.
d. Right angle turn. With rated load on the forks, forklift positioned perpendicular to a wall, and with the front of the load against the wall, the truck shall be able to back up and make a complete right angle turn (ending up parallel to the wall) within 200 inches.
e. Travel speed. While carrying the rated load on the forks, the travel speed shall be 7 miles per hour (mph) minimum (not to exceed 15 mph), in both forward and reverse. The vehicle speed shall be governed so that the maximum speed for safe operations is not exceeded.
f. Slope ascension. With the truck facing up the slope and with rated load on the forks, the truck shall be able to accelerate up a 15% (8.5 degree) slope from a dead stop.
g. Upright tilt. With no load on the truck, the forks shall have at least 5.0 degrees of forward tilt, and at least 8.0 degrees of rear tilt.
h. Upright height (collapsed mast height) and Maximum fork lift height. With no load on the forks and the forks on the ground, the maximum upright height shall be 92 inches. The maximum fork “lift height” shall be at least 184 inches with rated load on the forks and the forks horizontal.
i. Ground clearance. With rated load on the forks and mast vertical, ground clearance beneath the mast shall be at least 4.0 inches. With no load on the forks, the ground clearance of the truck other than the mast and axles shall be at least 5.0 inches.
j. Load drift. With hydraulic fluid at normal operating temperature, the lift assembly shall hold rated load at maximum lift height for at least 10.0 minutes with not more than 1-¾ inch of vertical drift, and not more than 1.0 degree of rotational drift.
k. Stability. The forklift shall meet the ANSI/ITSDF B56.1 "forward stacking", "forward travel", "lateral stacking", and "lateral travel" stability requirements.
l. Noise limits. Maximum allowable noise level shall not exceed 84dB(A) at the operator's ear. If the noise level at the operator's ear exceeds 84 dB(A) (85 to 92 dB(A)), a permanent warning decal or placard shall be installed inside the cab clearly visible to the operator. Noise level limits that exceed 92dB(A) are not acceptable.

2.21 Painting. The forklift shall be primed and painted using standard commercial practices. The final paint shall be a standard, commercially available polyurethane. The paint finish shall show no evidence of uneven application, curtains, runs snags, orange peel, lack of adhesion, or other defects. The color shall be the manufacturer’s standard color. If any components on the truck are not painted, the components shall have a commercially available, protective coating (such as zinc plating). When specified (see 6.3), the manufacturer shall make color 24052 of FED-STD-595 available.

2.22 Identification plates. Corrosion resistant identification plate shall be provided and mounted on the vehicle. The identification plate shall contain the following information:

Nomenclature:
Make and Model:
Manufacturers Serial Number (VIN):
Registration Number:

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National Stock Number (NSN):
Vehicle Curb Weight (lbs):
Payload, Maximum (lbs):
Gross Weight, Maximum (lbs):
Date of Delivery:
Warranty (months/miles):
Contract Number:
Ship to _____ Mark for _____
U S Property

2.23 Walkway coating. Floor plates and step surfaces shall be coated with a nonslip coating compound, or be furnished with a nonslip metal or treadplate surface.

2.24 Lifting and tiedown provisions. There shall be permanently installed lifting devices that enable the forklift to be lifted in its normal travel configuration, and tiedown devices that enable the forklift to be tied down to the floor of the transportation media. All devices shall have an inside diameter of 3.0 inches and minimum structural safety factor of 2:1, based on static load. The notation "LIFT HERE" and "TIEDOWN" shall be stenciled, in black, near each lifting and tiedown device.

2.25 Electromagnetic interference (EMI). The forklift shall comply with the EMI requirements shown in table I.

TABLE I. Electromagnetic interference (EMI)

Name	Description	Parameters
SAE J551-1	Compatibility	N/A
CISPR 12 & CISPR 25	Emissions	(Frequency Range: 30 to 1000 MHz)
SAE J551-11	Susceptibility	(Frequency Range: 100 kHz to 18 GHz) 20 V/m (100 kHz to 200 MHz) 50 V/m (above 200 MHz)

2.26 Highway transportability. The truck, when loaded on a semitrailer, shall be within the highway permit limits for all states.

2.27 Backup alarm. An audible backup alarm shall be furnished. The alarm shall automatically activate when the transmission selector is placed in reverse.

2.28 Flashing light. When specified (see 6.3), a flashing or strobe light shall be installed on the vehicle. The light shall not increase the overall height of the vehicle.

2.29 Workmanship. The truck shall withstand any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. Riveted and bolted connections shall be adequately designed. Welding shall be performed by a certified welder.

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3. REGULATORY REQUIREMENTS The offeror/contractor is encouraged to use recovered materials to the maximum extent practical, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the forklift.

4. PRODUCT CONFORMANCE

4.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5. PACKAGING. Preservation, packing and marking shall be as specified in the contract or order.

5.1. Preservation, Packaging, Labeling and Marking. Unless otherwise specified (see 6.3), the preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc. This involves shipment from the supply source to the receiving activity, and reshipment from the receiving activity. The preservation, packaging, and packing shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities, labeling and marking shall be as specified in the contract and/or order.

6. NOTES

6.1 Intended use. This commercial item description covers the general requirements for a diesel engine driven, front wheel drive, rear wheel steer, pneumatic tired, 12,000 pound capacity, forklift truck. The forklift is intended for handling cargo in and around warehouses, loading platforms, docks and on paved or other hard surfaces.

6.2 Sources of Documents.

6.2.1 Government documents. Copies of specifications, standards, and handbooks are available from http://assist.daps.dla.mil/ or the Standard Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.2.2 ITSDF Standards. Copies of ITSDF standards may be obtained from www.itsdf.org or the Industrial Truck Standards Development Foundation, Suite 460, 1750 K Street NW, Washington, DC 20006.

6.2.3 SAE Standards. Copies of SAE standards may be obtained from www.sae.org or the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.

6.2.4 TRA Standards. Copies of TRA standards may be obtained from www.us-tra.org or the Tire and Rim Association, 175 Montrose West, Suite 150, Copley, OH 44321.

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6.2.5 UL Standards. Copies of UL standards may be obtained from www.ul.com or the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096.

6.3 Ordering data. Procurement documents should specify the following:

a. Number and date of this CID
b. Type E, if required (see 1.1 and 2.1)
c. Additional battery (see 2.3)
d. Battery charger (see 2.3)
e. Optional forks, if required (see 2.6)
f. Enclosed Cab (see 2.18)
g. When final color shall be other than as specified (see 2.21)
h. Flashing light (see 2.28)
i. Packaging requirements, if different (see 5.1).

6.4 Keywords

Front wheel drive
Hydraulic
Rear-wheel steer

MILITARY INTEREST

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 84

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